PRIMARY STRUCTRE DESIGN FOR A FORMULA STUDENT VEHICLE

Formula Student FEUP 03

Department of Chassis

Department Leader:

Miguel Monteiro

Department Members Responsible:

Henrique Oliveira
Íris Botelho
Margarida Guerreiro
Matilde Rocha
Vasco Macedo

Reviewed by:

Miguel Monteiro

2025-2027

Contents

1 Conceptual Phase **1**
1.1 Non-Triangulated Suspension Hardpoints . . . 1
1.1.1 New Rear Frame iteration . . . 1
1.1.2 Validation . . . 3
1.2 FBHS Influence Study . . . 8

2 Design **9**
2.1 Frame . . . 9
2.1.1 Geometry . . . 9
2.1.2 Bulkheads . . . 32
2.1.3 Torsional Stiffness . . . 36
2.1.4 Local and Torsional Stiffness Optimization . . . 43
2.2 Primary Mounting . . . 50
2.2.1 Accumulator Mounting . . . 50
2.2.2 Lap Belt Mounting . . . 60
2.2.3 Steering Rack Mounts . . . 69
2.2.4 Steering Column Mount . . . 71
2.2.5 Front & Rear Bulkhead Mounts . . . 76
2.3 Suspension Mounts . . . 79
2.3.1 Wishbones Mounts . . . 80
2.3.2 Front Rocker Mount . . . 85
2.3.3 Rear Rocker Mount . . . 86
2.3.4 Front Damper Mount . . . 87
2.3.5 Rear Damper Mounting . . . 87
2.4 Secondary Mounting . . . 88
2.4.1 General Design Methodology . . . 88
2.4.2 Cooling Pump . . . 88
2.4.3 Dashboard . . . 89
2.4.4 LiDAR . . . 90
2.4.5 Pressure Cap . . . 91



Figure 2.14: FR [25-12-02-00]

FR [25-12-03-00] & [01]
A major update aimed at defining a first concrete tube layout to allow the vibrations subsystem to begin analysis.

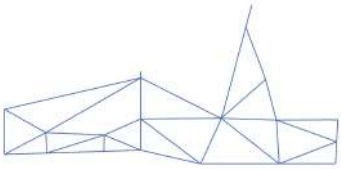


Figure 2.15: FR [25-12-03-00] wireframe

- **Front Bulkhead Support Structure (FBHSS):** The FBHSS was redesigned with a multi-part upper member and a continuous lower member, as illustrated in the figure 2.16.

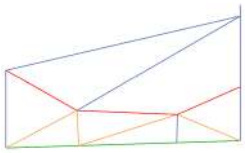


Figure 2.16: FBHSS: Upper member (red), Diagonal member (orange), Lower member (green)

- **Main Hoop (MH):** Out of plane bends implemented in the MH, to further adjust

Furthermore, it can be observed that as the modulus increases, the response tends toward an ideal rigid body behavior which would reproduce inaccurate values for the torsional stiffness of the overall system.

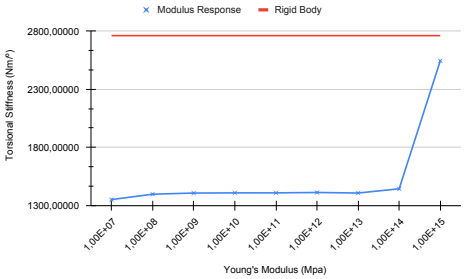


Figure 2.46: Sensitivity of Torsional Stiffness to Suspension Arm Modulus

The deformation pattern of the model, corresponding to the selected modulus value, is also presented in Figure 2.47. This visualization illustrates the structural response and displacement distribution under the applied torsional load.

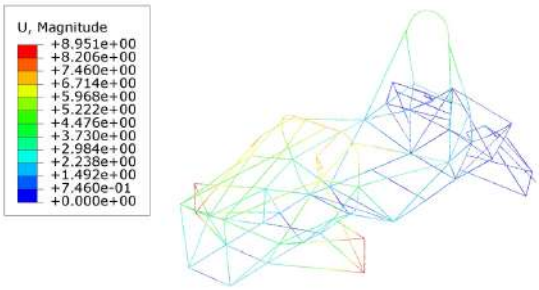
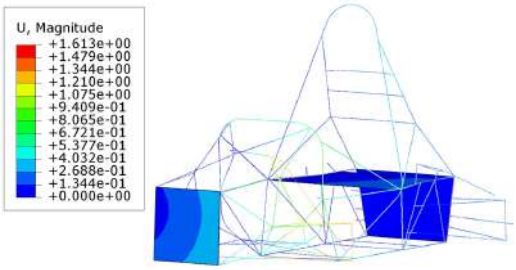
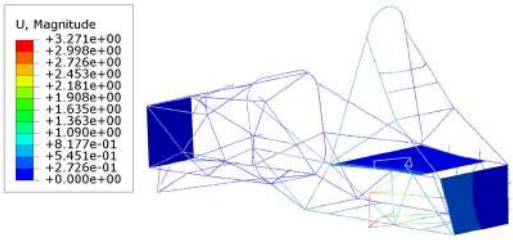


Figure 2.47: Torsional Displacement Gradient

The selected configuration, updated to ensure compliance with the regulations, is illustrated in Figure 2.53 alongside its corresponding tube cross-sections. The shaded values in the table are detailed through their displacement gradients in the subsequent figures. Finally, the torsional stiffness simulation yielded 1480 Nm/deg for the isolated frame. While this value is below the 1500 Nm/deg target, the inclusion of the structural shelf improves the overall torsional stiffness to 1569 Nm/deg, thereby meeting the design goal.



(a) Front axle displacement gradient (Case 25)



(b) Rear axle displacement gradient (Case 26)

Figure 2.54: Displacement gradient analysis for the critical load cases identified in the table

Table 2.11: Geometric properties of the weld group for stress analysis.

Property	Value
Weld Thickness (t)	3.00 mm
Mounting Angle (θ)	67.5°
Weld Group Area (A)	322.80 mm^2
Centroid Offset - x axis (L_x)	20.25 mm
Centroid Offset - y axis (L_y)	−3.38 mm
Centroid Offset - z axis (L_z)	0.00 mm
Moment of Inertia (I_y)	147 212.28 mm^4
Moment of Inertia (I_z)	29 770.87 mm^4
Distance to Extreme Fiber (c_y)	19.29 mm
Distance to Extreme Fiber (c_z)	27.00 mm

Vertical Load

$$F = 11\,183.4\,\text{N} \tag{2.25}$$

$$N = F \cdot \cos(\theta) = -4\,279.7\,\text{N} \tag{2.26}$$

$$V = F \cdot \sin(\theta) = -10\,332.11\,\text{N} \tag{2.27}$$

$$M_z = N \cdot L_y - V \cdot L_x = 223\,690.61\,\text{N} \cdot \text{mm} \tag{2.28}$$

$$\sigma(z) = \frac{M_z \cdot c_y}{I_z} + \frac{N}{A} = 157.81\,\text{MPa} \tag{2.29}$$

$$\tau(\text{avg}) = \frac{V}{A} = 31.05\,\text{MPa} \tag{2.30}$$

$$\sigma(\text{VM}) = \sqrt{\sigma(z)^2 + 3 \cdot \tau(\text{avg})^2} = 166.7\,\text{MPa} < \sigma_y \tag{2.31}$$

Lateral Load

$$F = 22\,366.8\,\text{N} \tag{2.32}$$

$$N = F \cdot \sin(\theta) = -20\,664.23\,\text{N} \tag{2.33}$$

$$V = F \cdot \cos(\theta) = 8\,559.4\,\text{N} \tag{2.34}$$

$$M_z = N \cdot L_y - V \cdot L_x = -103\,482.75\,\text{N} \cdot \text{mm} \tag{2.35}$$

$$\sigma(z) = \frac{M_z \cdot c_y}{I_z} + \frac{N}{A} = 129.15\,\text{MPa} \tag{2.36}$$